

Install from source.

01

One verified script prepares, builds, checks, and starts the local desktop app.

BEFORE YOU BEGIN

Use Terminal on Mac or PowerShell on Windows as a normal user. The first setup needs internet access for application dependencies and a private Node.js runtime when a compatible system copy is unavailable.

Mac - Terminal

```
git clone https://github.com/adammalin/Map-Maker-Studio.git \  
"$HOME/Map-Maker-Studio"  
cd "$HOME/Map-Maker-Studio"  
/bin/zsh scripts/setup-macos.zsh
```

Windows - PowerShell

```
git clone https://github.com/adammalin/Map-Maker-Studio.git \  
"$env:USERPROFILE\Map-Maker-Studio"  
Set-Location "$env:USERPROFILE\Map-Maker-Studio"  
powershell.exe -NoLogo -NoProfile -ExecutionPolicy Bypass \  
-File ".\scripts\setup-windows.ps1"
```

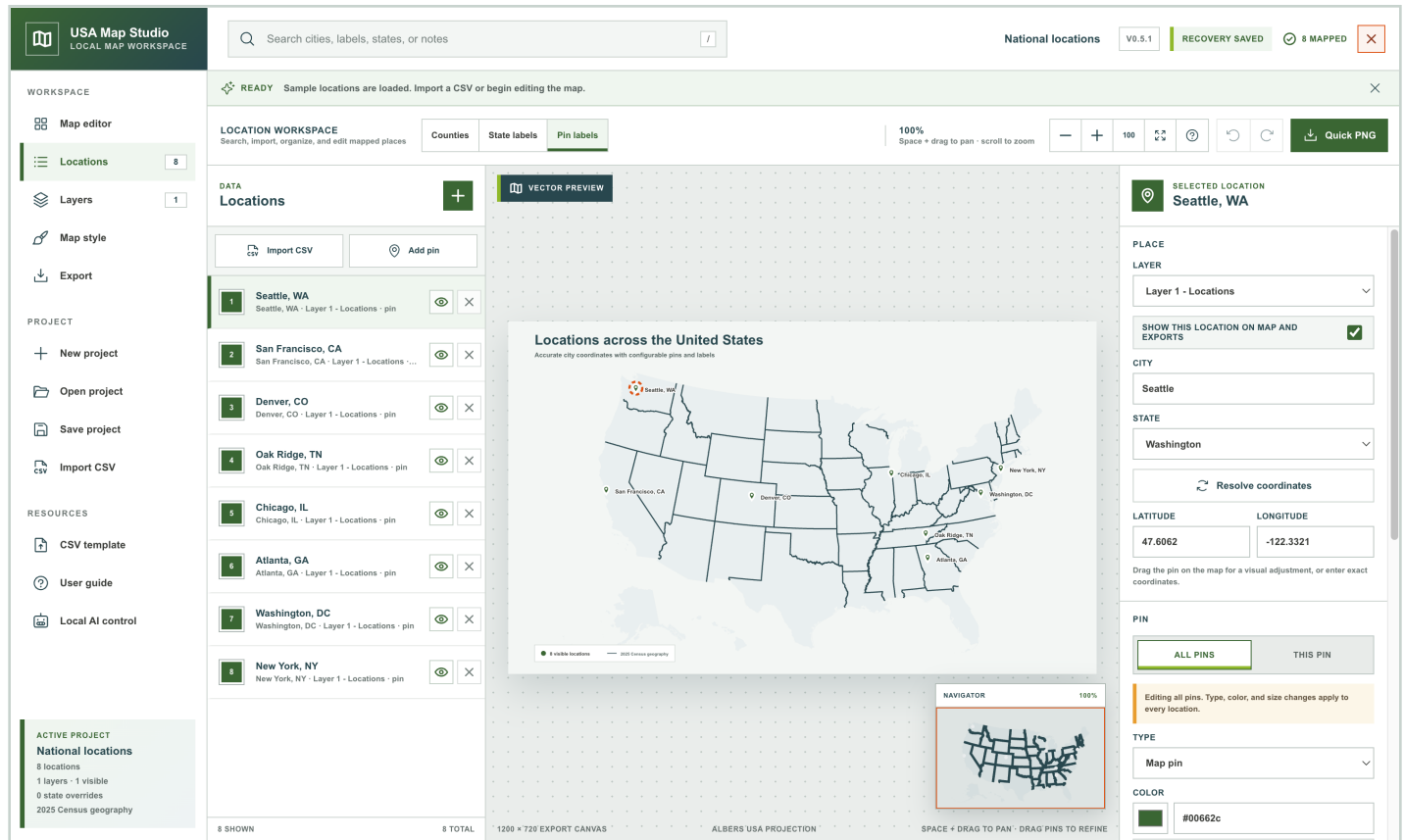
WHAT SETUP DOES

When rerun from a clean main Git checkout, setup first fast-forwards origin/main so it cannot quietly rebuild an old clone. Local changes and other branches are never overwritten. The script then installs exact dependencies, builds the Electron interface, and runs a hidden map/UI smoke check. It does not create a DMG, PKG, MSI, or EXE installer, disable Gatekeeper, or make system-wide changes. Later, double-click Start-USA-Map-Studio.command on Mac or Start-USA-Map-Studio.cmd on Windows.

Build the map visually.

02

A focused canvas, named layers, location list, and inspector keep every change visible.



1 MAP EDITOR

Select pins on the largest canvas and refine coordinates, labels, symbols, and colors.

2 LOCATIONS

Choose All pins (default) or This pin, then search, import, hide, show, and organize places.

3 LAYERS

Name, reorder, show, hide, and count separate location groups. Set one shared pin style.

4 MAP STYLE

Control fills, boundaries, county lines, label halo, abbreviations, and legend.

MAP CONTROLS AND SHORTCUTS

Hold Space and drag anywhere to pan; drag a pin without Space to move it. Scroll at the pointer or use +/- to zoom; 0 Fits and 1 returns to 100%. The bottom-right Navigator shows and recenters the visible region. Project keys: Cmd/Ctrl+S Save, Cmd/Ctrl+O Open, Cmd/Ctrl+N New, and Cmd/Ctrl+Z Undo. Cmd/Ctrl+Shift+Z or Ctrl+Y Redo; / searches; ? opens the keyboard reference.

Bring data in. Keep it whole.

03

Flexible columns feed a versioned project file that can be saved and reopened.

MINIMUM CSV

Two columns are enough

```
city,state
Oak Ridge,TN
Seattle,WA
```

Or supply exact coordinates

```
city,state,latitude,longitude
Denver,CO,39.7392,-104.9903
```

OPTIONAL VISUAL FIELDS

- **Visibility and labels**
visible hides or shows the complete location; show_label, label_color, and label_position control only its label.
- **Pins**
All pins is the default editing scope. Switch to This pin for one location. Type, color, size, and custom SVG choices follow that scope.
- **Additional data**
notes plus any custom columns. Unknown columns stay in customData and return with the project.
- **Choose the target layer**
The import summary separates ready rows from issues. Choose a named layer, then Add or Replace target layer. Other layers are untouched.
- **Save and autosave**
Save project chooses a .usmap.json path. Every later project change atomically updates that file and an internal recovery JSON.
- **Open project**
Opening validates the schema and required fields before replacing the canvas. Unsupported or unrelated JSON is rejected.

COORDINATE RULE

When latitude and longitude are blank, the app matches city and state to one of 32,350 representative points from the 2025 U.S. Census National Places Gazetteer. If both coordinates are supplied, the app preserves them. An unresolved place stays out of the map and appears in the import issue list so it cannot be plotted silently at the wrong location.

Export editable slide objects.

04

SVG, PNG, PowerPoint, and project JSON share one composition.

1 SVG

Scalable vector artwork with a named group for each visible location layer.

2 PNG

A 2400 x 1440 raster image for documents, email, and quick review.

3 PPTX

Editable states, text, pins, and legend using the visible pin size and viewport.

4 JSON

The complete editable project, including embedded custom pin artwork. Reopen it to continue work.

GOOD WORKING SEQUENCE

- 1 Save the project once**
Choose a .usmap.json path. Later project edits autosave to that file; a recovery JSON is restored on relaunch.
- 2 Review the visible map**
Set layers, pin size, and zoom/pan exactly as needed; SVG, PNG, and PowerPoint reproduce that view.
- 3 Keep source and output together**
Store the CSV and project JSON beside exported artwork when the map must be reproduced later.

TROUBLESHOOTING

Setup fails: rerun the setup script; it repairs dependencies and rebuilds safely. A city is unresolved: add exact coordinates or use the official Census place name. Hide a city: use its eye control; the row and JSON data remain. Labels overlap: move the label left, right, above, or below. Autosave fails: keep the app open, check the status chip, and use Save project to choose a writable location.

LOCAL DATA BOUNDARY

Normal editing, place lookup, map rendering, and export run locally. USA Map Studio does not send CSV or project content to a service. Files are written only to locations you choose. The bundled map and place data come from U.S. Census Bureau 2025 Cartographic Boundary and Gazetteer files.

Let AI prepare. You decide.

05

ChatGPT desktop, Codex, and compatible local clients can stage map changes for review.

1 KEEP THE APP OPEN

The setup script registers `usa_map_studio`. Restart the AI client, open USA Map Studio, then use `/mcp` to confirm the connection.

2 ASK FOR A DRAFT

The AI reads the named open project, uses its current timestamp, and stages one proposal. No map or saved file changes yet.

3 REVIEW IN THE APP

Compare Before and After. Apply or Reject. After Apply, check the canvas and autosave status; Save is only needed to choose a file path.

MANUAL CONNECTION COMMANDS

Register or repair

```
npm run mcp:install
# Restart ChatGPT desktop or Codex, then use /mcp
```

Remove managed connection

```
npm run mcp:remove
# Existing unrelated MCP settings remain in place
```

WHAT THE AI CAN DO

- **Read**
Check app status, read the project, list ordered layers or locations, and validate project JSON.
- **Prepare**
Create, rename, toggle, or remove layers; assign locations; target CSV imports; set one shared pin style; or stage other map changes.
- **Cannot bypass review**
Write tools create one pending proposal. They cannot click Apply or write an export. After you Apply, the app's normal autosave behavior takes over.
- **Stale changes stop**
If the working map changes after the AI reads it, the proposal must be rebuilt from the current project.

PRIVACY BOUNDARY

The bridge listens only on this computer and uses a new random token for each app launch. Project data stays local until a read tool returns it to the connected AI conversation. Use only an AI client approved for the map content. ChatGPT web cannot directly reach this private desktop bridge; that would require a separate hosted plugin.

Import a pin. Reuse the palette.

06

SVG artwork stays with the project, and named ORNL swatches stay one click away.

CUSTOM SVG PIN WORKFLOW

1 IMPORT

Choose All pins or This pin, select Import custom SVG pin, and open a vector-only .svg file up to 500 KB.

2 SHARE

Keep All pins selected to guarantee the same SVG, color, and size everywhere; use This pin for an exception.

3 SAVE

The sanitized SVG is stored inside .usmap.json. Move or share that one project file without losing the pin artwork.

PRIMARY



ORNL Green
#00662C



Hale Navy
#00454D



Graphite
#DBDCDB



Polar
#FFFFFF



Dark Matter
#373A36

SECONDARY



Energy
#7DBA00



Mist
#8BFEBF



Biome
#00B38F



Aqua
#00BDB5



Infinity
#006BA6



Hydro
#005776

ACCENT



Forge
#FF9E1B



Spark
#FE5000



Plasma
#B50094



Pulsar
#4E008E

SAFE BY DESIGN

The importer keeps safe Illustrator gradient and stroke styling while removing scripts, event handlers, external references, embedded raster images, unsafe styles, and unsupported elements. ORNL swatches are a draft aid: check contrast, use accents sparingly, and complete normal communications review before release.